

Exercici 1

A partir dels documents adjunts (estructura_dades i dades_introduir), importa les dues taules. Mostra les característiques principals de l'esquema creat i explica les diferents taules i variables que existeixen. Assegura't d'incloure un diagrama que il·lustri la relació entre les diferents taules i variables.

```

5      -- Creamos la tabla company
6      CREATE TABLE IF NOT EXISTS company (
7          id VARCHAR(15) PRIMARY KEY,
8          company_name VARCHAR(255),
9          phone VARCHAR(15),
10         email VARCHAR(100),
11         country VARCHAR(100),
12         website VARCHAR(255)
13     );
14
15
16     -- Creamos la tabla transaction
17     CREATE TABLE IF NOT EXISTS transaction (
18         id VARCHAR(255) PRIMARY KEY,
19         credit_card_id VARCHAR(15) REFERENCES credit_card(id),
20         company_id VARCHAR(20),
21         user_id INT REFERENCES user(id),
22         lat FLOAT,
23         longitude FLOAT,
24         timestamp TIMESTAMP,
25         amount DECIMAL(10, 2),
26         declined BOOLEAN,
27         FOREIGN KEY (company_id) REFERENCES company(id)

```

#	Time	Action	Message
5143	13:20:34	DROP DATABASE `transactions`	10 row(s) affected
5144	13:30:22	CREATE DATABASE IF NOT EXISTS transactions	1 row(s) affected
5145	13:30:22	USE transactions	0 row(s) affected
5146	13:30:22	CREATE TABLE IF NOT EXISTS company (id VARCHAR(15) PRIMARY KEY, comp...	0 row(s) affected
5147	13:30:22	CREATE TABLE IF NOT EXISTS transaction (id VARCHAR(255) PRIMARY KEY, cr...	0 row(s) affected

The screenshot shows a SQL IDE with a list of 21 INSERT statements for a 'transaction' table. Each statement includes columns: id, credit_card_id, company_id, user_id, lat, longitude, timestamp, amount, and declined. The values are represented by hex strings. Below the statements is an 'Output' pane showing 'Action Output' with a table of execution results.

#	Time	Action	Message
200205	13:49:30	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, am...	1 row(s) affected
200206	13:49:30	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, am...	1 row(s) affected
200207	13:49:30	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, am...	1 row(s) affected
200208	13:49:30	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, am...	1 row(s) affected
200209	13:49:30	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, am...	1 row(s) affected

Se ha creado una base de datos llamada “transactions

Dentro de las bases datos se ha creado dos tablas:

1- Company:

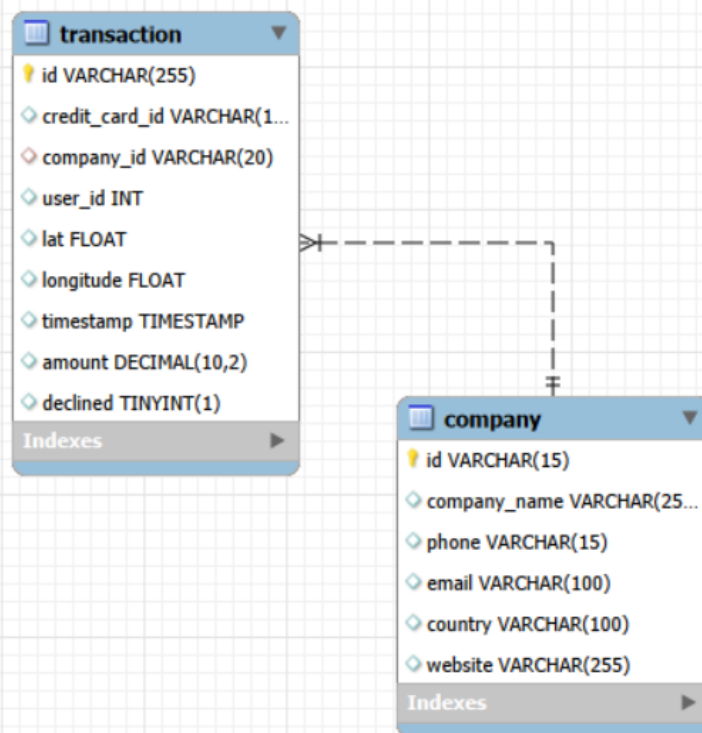
Nos proporciona informaci ó n de datos de las compa ñ í as que se identifica cada una con un primary Key, un numero identificativo ú nico para cada una.

2- transaction

a. Nos da informaci ó n de las transacciones realizadas identificadas por un primare key, ú nico para cada una. Tambi é n encontramos claves for á neas, como:

- i. Credit_card_id
- ii. Company_id
- iii. User_id
- iv. Las tablas est á n relacionadas por el company_id, en una relaci ó n 1 a N
(FOREIGN KEY (company_id) REFERENCES company(id))

1. Una empresa tiene varias transacciones, y una transacci ó n tiene una sola empresa.



Exercici 2

Utilitzant JOIN realitzar à s les seg ü ents consultes:

Nota general para todos los ejercicios:

*USE transactions; se ha puesto al inicio de cada ejercicio, al hacerse por separado. Cuando se han juntado ejercicios se ha dejada uno al inicio del Script. Lo ponemos para asegurarse que usamos esa base de datos, ya que tenemos varias y para hacer los ejercicios hemos utilizado un script por cada.

Aunque no aparezca en los pantallazos, en el script enviado deber í a aparecer DROP TABLE IF EXISTS Y DROP VIEW IF EXIST,

- Llistat dels països que estan generant vendes.

```
1 • USE transactions;
2
3 • SELECT DISTINCT c.country
4 FROM company c
5 INNER JOIN transaction t
6 ON c.id = t.company_id
7 WHERE declined = 0
8 Order by c.country ;
9
```

Result Grid | Filter Rows: | Export: |

country
Australia
Belgium
Canada
China
France
Germany
Ireland
Italy
Netherlands

- Des de quants països es generen les vendes.

```
10
11 • SELECT COUNT(DISTINCT c.country) AS paises_con_ventas
12 FROM company c
13 INNER JOIN transaction t
14 ON c.id = t.company_id
15 WHERE declined = 0;
16
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

paises_con_ventas
15

- Identifica la companyia amb la mitjana m é s gran de vendes.

```

7 • SELECT
8     c.company_name,
9     c.country,
10    AVG(t.amount) AS media_amount
11 FROM company c
12 INNER JOIN transaction t
13 ON c.id = t.company_id
14 GROUP BY c.id, c.company_name
15 ORDER BY media_amount DESC
16 LIMIT 1;
17
18

```

company_name	country	media_amount
Ac Fermentum Incorporated	Germany	284.867160

Exercici 3

Utilitzant nom é s subconsultes (sense utilitzar JOIN):

- Mostra totes les transaccions realitzades per empreses d'Alemanya.

```

1 • USE transactions;
2
3 • SELECT t.company_id, t.amount
4 FROM transaction t
5 WHERE declined = 0
6 AND EXISTS
7     (SELECT id
8      FROM company c
9      WHERE c.id = t.company_id
10      AND c.country = 'GERMANY'
11     );
12
13
14

```

company_id	amount
b-2222	426.36
b-2222	316.90
b-2222	453.04
b-2222	263.14
b-2222	442.01

- Llista les empreses que han realitzat transaccions per un amount superior a la mitjana de totes les transaccions.

```

15 • SELECT *
16 FROM company c
17 WHERE EXISTS
18 (
19     SELECT t. company_id
20     FROM transaction t
21     WHERE t.declined = 0
22     AND t.amount > (
23         SELECT AVG (amount)
24         FROM `transaction`
25     WHERE declined = 0)
26 );

```

Result Grid

	id	company_name	phone	email	country	website
▶	b-2222	Ac Fermentum Incorporated	06 85 56 52 33	donec.porttitor.tellus@yahoo.net	Germany	https://instagram.com/site
	b-2226	Magna A Neque Industries	04 14 44 64 62	risus.donec.nibh@icloud.org	Australia	https://whatsapp.com/group/9
	b-2230	Fusce Corp.	08 14 97 58 85	risus@protonmail.edu	United States	https://pinterest.com/sub/cars
	b-2234	Convallis In Incorporated	06 66 57 29 50	mauris.ut@aol.couk	Germany	https://cnn.com/user/110
	b-2238	Ante Iaculis Nec Foundation	08 23 04 99 53	sed.dictum.proin@outlook.ca	New Zealand	https://netflix.com/settings
	b-2242	Donec Ltd	01 25 51 37 37	at.iaculis@hotmail.couk	Norway	https://nytimes.com/user/110
	b-2246	Sed Nunc Ltd	02 62 64 73 48	nibh@yahoo.org	United Kingdom	https://cnn.com/one
	b-2250	Amet Nulla Donec Corporation	07 15 25 14 74	mattis.integer.eu@protonmail.net	Italy	https://netflix.com/sub/cars
	b-2254	Nascetur Ridiculus Mus Inc.	06 26 87 61 84	suspendisse.dui@icloud.net	United States	https://ebay.com/sub

- Eliminarian del sistema les empreses que no tenen transaccions registrades, entrega el llistat d'aquestes empreses.
 - Antes de borrar compruebo si existe empresas in transacciones, pero me sale que no hay, con lo cual no elimino nada.

[illegible]

Exercici 4

La teva tasca és dissenyar i crear una taula anomenada "credit_card" que emmagatzemi detalls crucials sobre les targetes de crèdit. La nova taula ha de ser capaç d'identificar de manera única cada targeta i establir una relació adequada amb les altres dues taules ("transaction" i "company"). Després de crear la taula serà necessari que ingressis la informació del document denominat "dades_introduir_credit". Recorda mostrar el diagrama i realitzar una breu descripció d'aquest.

Antes de cada tabla creamos:

DROP VIEW IF EXISTS “nombre_tabla”

```
1  USE transactions;
2
3  • CREATE TABLE IF NOT EXISTS credit_card
4
5  (id VARCHAR (100),
6   iban VARCHAR (100),
7   pan VARCHAR (6),
8   pin VARCHAR (6),
9   cvv VARCHAR (4),
10  expiring_date VARCHAR (8)
11  );
```

Result Grid						
Filter Rows:						
Export:  Wrap Cell Co						
id	iban	pan	pin	cvv	expiring_date	

```

4
5 (id VARCHAR (100),
6   iban VARCHAR (100),
7   pan VARCHAR (6),
8   pin VARCHAR (6),
9   cvv VARCHAR (4),
10  expiring_date VARCHAR (8)
11 );
12
13 • USE transactions;
14 • ALTER TABLE credit_card
15   MODIFY pan VARCHAR (50);
16
17 • DESCRIBE credit_card

```

Result Grid | Filter Rows: | Export: | Wrap

Field	Type	Null	Key	Default	Extra
id	varchar(100)	YES		NULL	
iban	varchar(100)	YES		NULL	
pan	varchar(50)	YES		NULL	
pin	varchar(6)	YES		NULL	
cvv	varchar(4)	YES		NULL	
expiring_date	varchar(8)	YES		NULL	

```

25 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CCU-3899', 'PT26105275356823705537218', '448 55418 98863 789', '5612', '564', '01/22/23');
26 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CCU-3106', 'AT684251637751136592', '349547146395283', '9733', '289', '01/27/24');
27 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CCU-3113', 'IE26LCGT47732173572752', '341834822877471', '9011', '287', '06/12/21');
28 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CCU-3120', 'RS72655766666166237144', '527646 533375 6577', '7658', '265', '01/16/21');
29 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CCU-3127', 'PT83533461438644342816864', '4716 443 46 4368', '8038', '924', '01/16/23');
30 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CCU-3134', 'BG23MYQ52668951824779', '5146 3453 9766 2168', '7260', '935', '08/24/25');
31 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CCU-3141', 'CH4437884777669672438', '3775 626726 45261', '2923', '338', '05/11/24');

```

Output

Action Output

#	Time	Action	Message	
5050	20:33:17	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES (CcS-9577, 'XX158914078594808633147086', '81104372718...	1 row(s) affected	0
5051	20:33:17	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES (CcS-9578, 'XX991539646456110567870254', '89998088230...	1 row(s) affected	0
5052	20:33:17	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES (CcS-9579, 'XX296383091587170202131236', '96900604686...	1 row(s) affected	0
5053	20:33:17	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES (CcS-9580, 'XX781258889851950806677358', '55411823644...	1 row(s) affected	0
5054	20:33:17	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES (CcS-9581, 'XX915670516405388124398147', '26243054701...	1 row(s) affected	0


```

18
19 • ALTER TABLE credit_card
20   ADD PRIMARY KEY (id);
21
22 • DESCRIBE credit_card;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

Field	Type	Null	Key	Default	Extra
id	varchar(100)	NO	PRI	NULL	
iban	varchar(100)	YES		NULL	
pan	varchar(50)	YES		NULL	
pin	varchar(6)	YES		NULL	
cvv	varchar(4)	YES		NULL	
expiring_date	varchar(8)	YES		NULL	

```

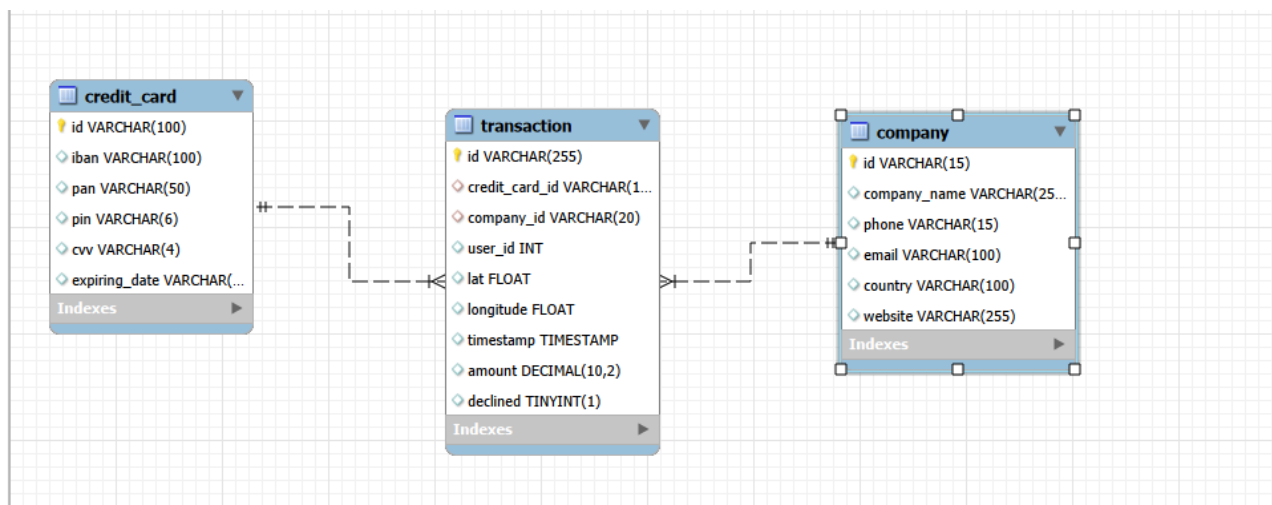
23 • USE transactions;
24
25 • ALTER TABLE `transaction`
26   ADD CONSTRAINT fk_credit_card
27   FOREIGN KEY (credit_card_id)
28   REFERENCES credit_card(id);

```

Output

Action Output

#	Time	Action	Message
5060	20:40:46	SELECT * FROM transactions.credit_card LIMIT 0, 1000	1000 row(s) returned
5061	20:42:07	ALTER TABLE credit_card ADD PRIMARY KEY (id)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
5062	20:42:17	DESCRIBE credit_card	6 row(s) returned
5063	20:42:48	USE transactions	0 row(s) affected
5064	20:42:48	ALTER TABLE `transaction` ADD CONSTRAINT fk_credit_card FOREIGN KEY (credit_card_id) REFERENCES credit_card(id)	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0



Exercici 5

El departament de Recursos Humans ha identificat un error en el número de compte associat a la targeta de crèdit amb ID CcU-2938. La informació que ha de mostrar-se per a aquest registre és: TR323456312213576817699999. Recorda mostrar que el canvi es va realitzar.

```
1 • USE transactions;
2   # encontrmos el registro a eliminar
3 • SELECT *
4   FROM credit_card
5   WHERE id = "CcU-2938";
6
7   #cambiamos registro en iban
8
9
10 • UPDATE credit_card
11     SET iban = 'TR323456312213576817699999'
12     WHERE id = "CcU-2938";
13
14   #Volvemos a buscar id para ver el cambio
15
16
17 • SELECT *
18     FROM credit_card
19     WHERE id = "CcU-2938";
```

result Grid					
Filter Rows:					
Edit:    Export/Import					
id	iban	pin	cvv	expiring_date	
CcU-2938	TR323456312213576817699999	3257	984	10/30/22	
NULL	NULL	NULL	NULL	NULL	

Exercici 6

En la taula "transaction" ingressa una nova transacció amb la següent informació:

Al ingresar los datos en la tabla transaction, se produce un error porque los valores de company_id y credit_card_id no existen previamente en las tablas company y credit_card.

Esto ocurre porque transaction contiene claves FK que dependen de esas tablas. Por tanto, antes de registrar una transacción, deben existir primero los datos de la empresa y de la tarjeta de crédito relacionadas.

Primero se introducen los datos en company y credit_card, y posteriormente se inserta la transacción en la tabla transaction.

```
1 • USE transactions;
2
3 • SELECT *
4   FROM company
5  WHERE id = 'b-9999';
```

< Result Grid | Filter Rows: | Edit: |

	id	company_name	phone	email	country	website
*	NULL	NULL	NULL	NULL	NULL	NULL

```
1 • USE transactions;
2
3 • SELECT *
4   FROM credit_card
5  WHERE id = 'CcU-9999';
6
```

< Result Grid | Filter Rows: | Edit: |

	id	iban	pan	pin	cvv	expiring_date
*	NULL	NULL	NULL	NULL	NULL	NULL

```
7 • USE transactions;
8
9 • INSERT INTO credit_card
10 (id, iban, pan, pin, cvv, expiring_date)
11 VALUES
12 ('CcU-9999',
13  'ES1234567890123456789012',
14  '4532123456789012',
15  '1234',
16  '456',
17  '12/28');
```

```

6
7 • USE transactions;
8 • INSERT INTO company
9   (id, company_name, phone, email, country, website)
10  VALUES
11  ('b-9999',
12   'Empresa Prueba S.L.',
13   '+34 931234567',
14   'info@empresaprueba.com',
15   'SPAIN',
16   'www.empresaprueba.com');
17
18 • SELECT *
19   FROM company
20  WHERE id = 'b-9999';
21

```

Result Grid		Filter Rows:		Edit:				Export/Import:			Wrap Cell Content:	
	id	company_name	phone	email	country	website						
▶	b-9999	Empresa Prueba S.L.	+34 931234567	info@empresasprueba.com	SPAIN	www.empresasprueba.com						
*	NULL	NULL	NULL	NULL	NULL	NULL						

```
1 #USE transactions;
2
3 #INSERT INTO `transaction` (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, amount, declined)
4 #VALUES ('108B1D1D-5B23-A76C-55EF-C568E49A99DD','CcU-9999','b-9999','9999','829.999','-117.999', NOW(), '111.11','0');
5
6 • USE transactions;
7
8 • Select *
9 FROM transaction
10 WHERE credit card id = 'CcU-9999';
```

[illegible]

Exercici 7

Des de recursos humans et sol • liciten eliminar la columna "pan" de la taula credit_card. Recordar mostrar el canvi realitzat.

```
1 • USE transactions;
2
3 • DESCRIBE credit_card;
4
5 • ALTER TABLE credit_card
6 DROP COLUMN pan;
```

Field	Type	Null	Key	Default	Extra
id	varchar(100)	NO	PRI	NULL	
iban	varchar(100)	YES		NULL	
pan	varchar(50)	YES		NULL	
pin	varchar(6)	YES		NULL	
cvv	varchar(4)	YES		NULL	
expiring_date	varchar(8)	YES		NULL	

```
2
3 • DESCRIBE credit_card;
4
5 • ALTER TABLE credit_card
6 DROP COLUMN pan;
```

Field	Type	Null	Key	Default	Extra
id	varchar(100)	NO	PRI	NULL	
iban	varchar(100)	YES		NULL	
pin	varchar(6)	YES		NULL	
cvv	varchar(4)	YES		NULL	
expiring_date	varchar(8)	YES		NULL	

Exercici 8

Descarrega els arxius CSV que trobaràs a l'apartat de recursos:

Estudia'ls i dissenya una base de dades amb un esquema d'estrella que contingui, almenys 4 taules de les quals puguis realitzar les següents consultes:
La taula de products.csv l'utilitzarem més endavant.

- o american_users.csv

Table: american_users

Columns:

id	varchar(15) PK
name	varchar(20)
surname	varchar(20)
phone	varchar(20)
email	varchar(50)
birth_date	varchar(50)
country	varchar(20)
city	varchar(20)
postal_code	varchar(20)
address	varchar(100)
signup_date	varchar(50)
user_segment	varchar(20)
income_band	varchar(20)

o

Limit to 1000 rows

1 • SELECT * FROM transactions.american_users;

Result Grid

	id	name	surname	phone	email	birth_date	country	city	postal_code	address	signup_date
▶	1	Zeus	Gamble	1-282-581-0551	interdum.enim@protonmail.edu	Nov 17, 1985	United States	New York	10001	348-7818 Sagittis St.	2014-04-21
	10	Robert	Mccarthy	(324) 746-6771	fermentum@protonmail.com	Apr 30, 1984	United States	San Jose	95101	P.O. Box 773	2013-10-15
	100	Melodie	Mclean	1-677-221-7152	risus.varius@google.ca	Sep 15, 1989	United States	San Jose	95101	Ap #644-8492 Sagittis St.	2015-04-20
	1006	Rbkocq	Swkmynuj	+38-186-8580	rbkocq.swkmynuj@example.com	Dec 2, 1989	United States	New York	10001	567 Swkmynuj St	2014-12-05
	101	Sarah	Beck	(358) 691-4345	vitae.risus@aol.couk	Apr 9, 1983	United States	Philadelphia	19101	665-9047 In	2014-08-11

```
1 • SELECT count(*) FROM transactions.american_users;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
count(*)			
1010			

o companies.csv

N1_ex_8_CREATE_TABLE american_users american_users companies x

Limit to 1000 rows

```
1 • SELECT * FROM transactions.companies;
```

company_id	company_name	phone	email	country	website	merchant_category	merchant_price_position
b-2230	Fusce Corp.	+29 657 646 4324	risus@protonmail.edu	United States	https://www.fusce-corp.com	Home	mid-market
b-2242	Donec Ltd	+56 897 341 8418	at.iaculis@hotmail.couk	Norway	https://www.donec-ltd.com	Home	mid-market
b-2246	Sed Nunc Ltd	+5 218 900 3193	nibh@yahoo.org	United Kingdom	https://www.sed-nunc-ltd.com	Home	mid-market
b-2258	Vestibulum Lorem PC	+27 788 560 5000	aenean.massa.integer@aol.net	Belgium	https://www.vestibulum-lorem-pc.com	Electronics	premium
b-2262	Gravida Sagittis LLP	+98 204 198 9702	turpis.vitae@google.ca	Sweden	https://www.gravida-sagittis-llp.com	Sports	premium
b-2274	Sed LLC	+3 536 122 8140	at@outlook.com	Belgium	https://www.sed-llc.com	Fashion	mid-market
b-2278	Arcu LLP	+53 776 813 5178	dui@aol.ca	Norway	https://www.arcu-llp.com	Home	mid-market

Administration		Schemas
Information		
Table: companies		
Columns:		
company_id	varchar(20)	PK
company_name	varchar(20)	
phone	varchar(20)	
email	varchar(100)	
country	varchar(20)	
website	varchar(150)	
merchant_category	varchar(20)	
merchant_price_position	varchar(20)	

N1_ex_8_CREATE_TABLE american_users american_users companies x

 Limit to 1000 rows

1 • `SELECT COUNT(*) FROM transactions.companies;`

<

Result Grid

 Filter Rows:

Export: 

Wrap Cell Content: 

	COUNT(*)
▶	57

o credit_cards.csv

N1_ex_8_CREATE_TABLE american_users american_users companies credit_cards

Limit to 1000 rows

1 • SELECT * FROM transactions.credit_cards;

	id	user_id	iban	pan	pin	cvv	track1	track2	expiring_date	card_type	Card_renewal_flag
▶	CcS-4857	276	ES415679416676968711699208	2314242385113924	1819	467	%B2314242385113924^LWCBUDLWCBUD^22...	%B2314242385113924=24101015183631647	09/25/28	debit	1
	CcS-4858	277	NL252053037609851262242244	6582720299715533	3964	817	%B6582720299715533^TIQMVITIQMVI^2404...	%B6582720299715533=24111011045462727	12/28/28	debit	0
	CcS-4859	278	SE720084318584364771497262	8861684536289642	4983	277	%B8861684536289642^COFBGDCOFBGD^280...	%B8861684536289642=25021017616653717	11/26/28	credit	0
	CcS-4860	279	ES165574078045314365283427	2481155515498459	6876	661	%B2481155515498459^TIUJTUTIUJTU^31040...	%B2481155515498459=26021015144143957	07/27/28	prepaid	0

Information

Table: credit_cards

Columns:

id	varchar(20) PK
user_id	varchar(20)
iban	varchar(50)
pan	varchar(50)
pin	varchar(50)
cvv	varchar(50)
track1	varchar(150)
track2	varchar(150)
expiring_date	varchar(20)
card_type	varchar(20)
Card_renewal_flag	varchar(20)

Limit to 1000 rows

1 • SELECT COUNT(*) FROM transactions.credit_cards;

	COUNT(*)
▶	5000

o european_users.csv

N1_ex_8_CREATE_TABLE american_users american_users companies credit_cards european_users

Limit to 1000 rows

1 • SELECT * FROM transactions.european_users;

id	name	surname	phone	email	birth_date	country	city	postal_code	address	signup_date	user_segment	income_band
1000	Amkjrjv	Qbulrxbp	+48-258-9936	amkjrjv.qbulrxbp@example.com	May 17, 1970	Germany	Stuttgart	70173	215 Qbulrxbp St	2015-10-28	occasional	low
1001	Nfvrjb	Oydaivbg	+94-121-2522	nfvrjb.oydaivbg@example.com	Mar 4, 1994	Germany	Cologne	50667	121 Oydaivbg St	2014-05-22	premium	very_high
1002	Ijbfmd	Jbddzhvp	+70-120-3668	ijbfmd.jbddzhvp@example.com	Sep 27, 2001	Germany	Munich	80331	412 Jbddzhvp St	2013-02-11	student	medium
1003	Uycig	Sfbdymzj	+58-123-6968	uydig.sfbdyzmj@example.com	Jan 20, 1981	Germany	Stuttgart	70173	735 Sfbdymzj St	2014-01-29	family	medium
1004	Yjqurq	Ojizvgqi	+77-944-2340	yjqurq.ojizvgqi@example.com	Jul 27, 1954	Germany	Munich	80331	685 Ojizvgqi St	2014-11-22	established	medium
1005	Mnlqtu	Glofegwk	+54-801-2627	mnlqtu.glofegwk@example.com	Nov 15, 1962	Portugal	Funchal	9000-001	8 Glofegwk St	2015-01-13	occasional	medium
1007	Ehcrjp	Xahkzrlm	+71-862-6101	ehcrjp.xahkzrlm@example.com	Nov 8, 1971	Spain	Sevilla	41001	205 Xahkzrlm St	2013-02-18	young_professional	low
1008	Securp	Faofvqfy	+76-822-2041	securp.faofvqfy@example.com	Nov 13, 1957	Sweden	Stockholm	111 20	535 Faofvqfy St	2014-11-28	established	medium

Table: **european_users**

Columns:

id	varchar(20) PK
name	varchar(20)
surname	varchar(20)
phone	varchar(20)
email	varchar(50)
birth_date	varchar(20)
country	varchar(20)
city	varchar(20)
postal_code	varchar(20)
address	varchar(50)
signup_date	varchar(20)
user_segment	varchar(50)
income_band	varchar(20)

Limit to 1000 rows

1 • SELECT COUNT(*) FROM transactions.european_users;

COUNT(*)
3990

○ transactions.csv

Table: transactions

Columns:

Column	Data Type
id	varchar(50) PK
card_id	varchar(50)
business_id	varchar(50)
timestamp	varchar(20)
amount	decimal(10,2)
declined	tinyint(1)
product_ids	varchar(50)
user_id	varchar(50)
lat	varchar(50)
longitude	varchar(50)
discount_amount	decimal(10,2)
tax_amount	decimal(10,2)
shipping_amount	decimal(10,2)
channel	varchar(50)
campaign_id	varchar(50)
device_type	varchar(50)
is_international	tinyint(1)
decline_reason	varchar(50)
distance_km	decimal(10,2)

○ users

Table: users

Columns:

Column	Data Type
id	varchar(100) PK
name	varchar(20)
surname	varchar(20)
phone	varchar(20)
email	varchar(100)
birth_date	varchar(20)
country	varchar(20)
city	varchar(20)
postal_code	varchar(10)
address	varchar(150)
signup_date	varchar(20)
user_segment	varchar(100)
income_band	varchar(100)

Al crear la relación entre tablas, hay problemas con la longitud del varchar en algunas tablas, durante el ejercicio se han hecho modificaciones para arreglarlo:

Modificamos varchar

ALTER TABLE companies

MODIFY COLUMN company_id VARCHAR(100);

ALTER TABLE users

MODIFY COLUMN id VARCHAR(100);

ALTER TABLE companies

MODIFY COLUMN company_name VARCHAR(200);

Exercici 10

Mostra la mitjana d'amount per IBAN de les targetes de cr è dit a la companyia Donec Ltd, utilitza almenys 2 taules.

```
3
4 • SELECT
5     cc.iban,
6     ROUND(AVG(t.amount), 2) AS media_amount
7 FROM transactions t
8
9 JOIN credit_cards cc
10     ON t.card_id = cc.id
11
12 JOIN companies c
13     ON t.business_id = c.company_id
14
15 WHERE c.company_name = 'Donec Ltd'
16 GROUP BY cc.iban;
```

<		
Result Grid		
Filter Rows:		
Export:  Wrap Cell Content: 		
	iban	media_amount
▶	CA901355773586771626108990	361.01
	US729212642816955651428791	142.56
	US899450803230933276819251	246.90
	US581097432668325541142297	154.01
	CA441226267105857473570574	250.51
	CA332171962352073015086775	203.95

Nivell 2

Exercici 1

Identifica els cinc dies que es va generar la quantitat més gran d'ingressos a l'empresa per vendes. Mostra la data de cada transacció juntament amb el total de les vendes.

```
3 • USE TRANSACTIONS;
4
5 • SELECT
6     DATE (TIMESTAMP) AS fecha,
7     ROUND(SUM(t.amount),2) AS total_amount
8 FROM transactions t
9 WHERE declined = 0
10 GROUP BY date(timestamp)
11 ORDER BY total_amount desc
12 Limit 5;
13
14
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content
	fecha	total_amount			
▶	2023-02-20	15728.40			
	2019-11-18	15447.89			
	2022-12-13	14944.00			
	2019-03-18	14896.47			
	2017-12-20	14367.96			

Exercici 2

Presenta el nom, tel·lèfon, país, data i amount, d'aquelles empreses que van realitzar transaccions amb un valor comprès entre 350 i 400 euros i en alguna d'aquestes dates: 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024. Ordena els resultats de major a menor quantitat.

```
7 • USE TRANSACTIONS;
8
9 • SELECT
10   c.company_name,
11   c.phone,
12   c.country,
13   DATE (t.timestamp) AS date,
14   Round(t.amount,2) As amount
15 FROM transactions t
16 JOIN company c
17 ON t.business_id = C.id
18 WHERE DATE(t.timestamp) IN ('2015-04-29', '2018-07-20', '2024-03-13')
19 AND declined = 0
20 AND t.amount BETWEEN 350 AND 450;
21
22
```

company_name	phone	country	date	amount
Pede Cum Ltd	07 62 26 48 38	Norway	2018-07-20	358.64
Orci Adipiscing Limited	03 18 00 77 81	United Kingdom	2018-07-20	414.62
Auctor Mauris Vel LLP	08 09 28 74 14	United States	2024-03-13	383.58
Lorem Ipsum Dolor Corp.	09 03 14 76 02	Sweden	2015-04-29	419.41
Tempor Diam Institute	05 01 94 80 87	Netherlands	2024-03-13	350.71
Viverra Donec Foundation	03 33 12 32 73	United Kingdom	2018-07-20	406.39
Netus Et Malesuada Ltd	02 55 43 68 46	Netherlands	2024-03-13	352.21
Fringilla LLC	08 29 15 93 57	New Zealand	2015-04-29	390.33
Mauris Institute	05 29 60 36 87	Sweden	2015-04-29	369.20
Auctor Mauris Vel LLP	08 09 28 74 14	United States	2018-07-20	423.34

Exercici 3

Necessitem optimitzar l'assignació dels recursos i dependrà de la capacitat operativa que es requereixi, per la qual cosa et demanen la informació sobre la quantitat de transaccions que realitzen les empreses, però el departament de recursos humans és exigent i vol un llistat de les empreses on especifiquis si tenen igual o més de 400 transaccions o menys.

```
7 • USE transactions;
8
9 • SELECT
10     c.company_name,
11     COUNT(t.id) AS Cantidad_transacciones,
12
13     CASE
14         WHEN COUNT(t.id) >= 400 then '400 o más transacciones'
15         ELSE 'MENOS DE 400 transacciones'
16     END AS capacidad_operativa
17
18 FROM transactions t
19 JOIN company c
20 ON c.id = t.business_id
21 GROUP BY c.company_name;
22
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	company_name	Cantidad_transacciones	capacidad_operativa
▶	Ac Fermentum Incorporated	2401	400 o más transacciones
	Magna A Neque Industries	410	400 o más transacciones
	Fusce Corp.	447	400 o más transacciones
	Convallis In Incorporated	1514	400 o más transacciones
	Ante Iaculis Nec Foundation	472	400 o más transacciones
	Donec Ltd	449	400 o más transacciones
	Sed Nunc Ltd	1541	400 o más transacciones
	Amet Nulla Donec Corporation	1511	400 o más transacciones

Exercici 4

Elimina de la taula transaction el registre amb ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de dades.

```
4 • USE transactions;
5
6 • DELETE FROM transaction
7 WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD';
8
9 • SELECT *
10 FROM transaction
11 WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD';
```

[illegible]

Exercici 5

La secció de màrqueting desitja tenir accés a informació específica per a realitzar anàlisi i estratègies efectives. S'ha sol·licitat crear una vista que proporcioni detalls clau sobre les companyies i les seves transaccions. Serà necessària que creïs una vista anomenada VistaMarketing que contingui la següent informació: Nom de la companyia. Telèfon de contacte. País de residència. Mitjana de compra realitzat per cada companyia. Presenta la vista creada, ordenant les dades de major a menor mitjana de compra.

```
6 • USE transactions;
7
8 • DROP VIEW IF EXISTS VistaMarketing;
9 • CREATE VIEW VistaMarketing AS
10 SELECT
11     c.company_name,
12     c.phone,
13     c.country,
14     ROUND(AVG (t.amount),2) AS media_compra
15 FROM `transactionS` t
16 JOIN company c
17 ON c.id = t.business_id
18 WHERE declined= 0
19 GROUP BY
20     c.id
21 ORDER BY media_compra DESC;
22
23 • SELECT *
24 FROM VistaMarketing
25 ORDER BY media_compra DESC;
```

company_name	phone	country	media_compra
Ac Fermentum Incorporated	06 85 56 52 33	Germany	330.71
Ut Semper Foundation	01 60 36 33 06	Sweden	307.30
Dolor Vitae Limited	06 53 60 43 60	France	306.21
Orci Adipiscing Limited	03 18 00 77 81	United Kingdom	305.87
Nunc In Foundation	03 79 81 18 92	Italy	305.67
Maecenas Malesuada Fringilla Inc.	09 38 53 76 61	Netherlands	304.49
Turpis Company	09 69 48 65 52	Netherlands	303.47

Nivell 3

Exercici 1

Crea una nova taula que reflecteixi l'estat de les targetes de cr è dit (CREATE VIEW --AS) basat en si les tres últimes transaccions han estat declinades aleshores és inactiu, case / grupu by-order bysi almenys una no és rebutjada aleshores és actiu. Partint d' aquesta taula respon:

Quantes targetes estan actives? Count(cards_id)

```

1  use transactions;
2  DROP VIEW IF EXISTS Estado_Tarjetas;
3  CREATE VIEW Estado_Tarjetas AS
4  WITH ultimas_transacciones AS (
5      SELECT
6          t.card_id,
7          t.declined,
8          t.timestamp,
9          ROW_NUMBER() OVER (
10             PARTITION BY t.card_id
11             ORDER BY t.timestamp DESC
12         ) AS numero_transacciones
13      FROM transactions t
14  )
15  SELECT
16      cc.id,
17      cc.user_id,
18      cc.iban,
19      CASE
20          WHEN COUNT(ut.card_id) = 3 AND SUM(ut.declined) = 3 THEN 'Inactiva'
21          ELSE 'Activa'
22      END AS estado_tarjeta
23  FROM credit_cards cc
24  LEFT JOIN ultimas_transacciones ut
25      ON cc.id = ut.card_id
26      AND ut.numero_transacciones <= 3
27  GROUP BY
28      cc.id,
29      cc.user_id,
30      cc.iban;
31
32  SELECT COUNT(*) AS tarjetas_activas
33  FROM Estado_Tarjetas
34  WHERE estado_tarjeta = 'Activa';
35

```

Result Grid

tarjetas_activas
5000